

SUMMARY

Dedicated Computer Science student with a strong focus on ethics and self-improvement. Experienced in developing AI/ML and IoT solutions, with a focus on resilient , high-quality code and implementing automated software development lifecycles. Proven ability to collaborate in fast-paced hackathons and deliver innovative solutions.

EDUCATION

- **University of Engineering and Management (UEM)**
B.Tech in Computer Science and Engineering; YGPA: 8.98 (1st Yr), 8.68 (2nd Yr)

Kolkata, India
2023 – Present
- **Don Bosco School, Bandel**
ISC (Grade 12): 85.5%; ICSE (Grade 10): 95.6%

Hooghly, India
2011 – 2023

AREAS OF INTEREST

OS, DBMS, OOPs, Software Engineering, AI/ML, Data Structures, Networking

SKILLS SUMMARY

- **Languages:** Python, Java, C, MySQL, JavaScript
- **Tools:** Git, Linux (Ubuntu/Kali), Flask, Tailwind CSS, Docker, Raspberry Pi
- **AI/ML:** Neural Networks,Big Data, Explainable AI (XAI), Computer Vision,TensorFlow,Keras, Mediapipe, Fuzzy Logic
- **Cloud/Other:** AWS, Google Cloud, CI/CD, Agile Methodologies

EXPERIENCE

- **IEDC UEM Kolkata**
IoT Researcher & Developer Intern

Kolkata, India
Aug 2025 – Present

◦ **Awarded Grade 'A':** Developed a predictive riverbank health model (**97% accuracy**) using Attention-Based Neural Networks and XAI.

◦ **MITTIE:** Architected an IoT ecosystem for soil monitoring to optimize crop yield via multi-parameter data streams.
- **CodeAlpha**
Machine Learning Intern

Remote
Jul 2025 – Aug 2025

◦ **XAI Risk Modeling:** Built transparent Credit Scoring (**96.5% acc.**) and Healthcare diagnostic models using LIME/SHAP for "Glass-Box" decision-making.
- **GAOTek Inc.**
Technical Staff Intern

New York, USA (Remote)
Jun 2024 – Aug 2024

◦ **Market Intelligence:** Leveraged Generative AI to automate Comparative Market Analysis (CMA) and technical product documentation.

PROJECTS & RESEARCH

- **MITTIE (IoT & AI Agritech):** Raspberry Pi-based system for real-time soil analysis and disease detection using voice-enabled chatbot integration. (Jan '26)
- **Coronary Artery Segmentation:** Glass-Box research using **U-Net++** (Dice Score: **0.77**) and Grad-CAM for medical diagnosis visual explainability. (2025)
- **Riverbank Health Prediction:** Implemented Attention-Based Neural Networks (**97% accuracy**) with Fuzzy Enhanced Feature Engineering for XAI 2.0. (2025)

RESEARCH & PUBLICATIONS

- **NOVA Publication 2025:** "Explainable Semantic Segmentation of Coronary Arteries" (Accepted)
- **ICDMIS 2025:** "Fuzzy Driven Engineering for Riverbank Health Prediction" (Accepted)
- **ICELTS 2024:** "Revolutionising Literacy by Introducing a Device for Literacy Improvement" (Accepted)

PROFESSIONAL CERTIFICATIONS

- **AI Developer Professional Certificate:** IBM Jun 2025
- **Google Cybersecurity Professional:** Google Jun 2025
- **Generative AI for Software Developers Specialization:** IBM Jun 2025
- **Applied Cryptography Specialization:** University of Colorado Oct 2025
- **Data Analysis and Visualization Foundations Specialization :** IBM Jun 2025

HONORS & EXTRACURRICULARS

- National Finalist (Top 10): AI for Good Fest 2026 (EMPI Business School-Delhi) for the MITTIE project; Represented team at Square Hacks (IIT Delhi) and ReGen 2.0 (NIT Manipur) for top-tier innovative solutions.
- First Prize: ML CANVAS 2.0 Poster Competition & Adcomsys Tech for excellence in machine learning visualization.
- Professional Artist,Athlete& Leadership: 5th-year certified in Painting; Represented school in District-level Football(Captain) and competitive Table Tennis, Organized 10 technical events